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Phishing Scams on Social Media: Analyzing the Impact and Effectiveness of Cyber Awareness Education

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Abstract

The danger of phishing to social media is a security-conception dilemma deriving from human judgment. This article aims to present the state-of-the-art knowledge of mechanisms of phishing in social networks and analyze the awareness training effectiveness in reducing susceptibility. The paper includes the analysis of 25 sources on the anatomy of phishing (Instagram-based and direct-message), vulnerabilities in online social networks, human and cognitive factors in social media phishing, phishing and social engineering, training in phishing awareness, machine-learning-based detection and adaptive security education. The sources were grouped into the following themes: factors of susceptibility to social-media phishing; effectiveness of awareness training; technical and other challenges of the current research. As a result of the analysis, the paper highlights the mixed evidence on training effectiveness, with interactive and adaptive simulators demonstrating increased detection rates in lab experiments, while traditional and embedded training failed in a field experiment with more than 19.5k participants. The study also reports four methodological limitations in the current literature, including the absence of platform-specific assessment, behavioral outcome consideration, retention measurements, and standardized evaluation protocols. Therefore, the article proposes a combined approach for the evaluation of security education effectiveness on the basis of pre-test/simulation/post-test/retention protocol and four scenario archetypes to measure precision, accuracy, recall, F1-score, and unsafe-action rate of social-media phishing. Finally, the paper notes that all quantitative results are taken from the summarized sources, while the study itself did not include the participant data collection.

Keywords: social media phishing; cyber awareness; social engineering; phishing susceptibility; security education; behaviours evaluation.

2. Introduction

Social media (SM) platforms have become a hub of communication, commerce, education, and professional networking. Their network effects, speed of information exchange, and relational trust mechanisms make these digital platforms simultaneously attractive and challenging security environments. The threat of phishing emerges through the tripartite relationship between the attacker, victim, and communication channel, and the multiplicity of attacking vectors and social engineering techniques [1]. Within the social media context, the recruiter's trusted persona is the lure that initiates the sequence of targeting actions. Nasution et al. demonstrate the use of fake web pages and interactive phishing on Instagram to establish contact and collect victim information for attack profiling [2]. Earlier research involving phishing campaigns among Indiana University students via email revealed the effectiveness of exploiting familiar personalities as the initiation of deception [3]. In particular, 72% of students fell under the phishing scheme's impact when the attacker's identity was a 'friend.' From the technical side, phishing on social media is a detection and selection behavior issue.

The detection inconsistency derives from the apparent contradiction between the phishing campaign's knowledge and the probability of becoming a victim of a well-structured social engineering attack. Perhaps, a technically competent individual would not be deceived by a phishing campaign as easily as a novice user, lured by an offer from a popular brand. Instead, users tend to fall for deception attempts that appeal

to their sense of individuality and belonging, such as messages from friends or the social media platform itself. In particular, the user-related factors include motivation, engagement, experience, trust, and competence [4], [5]. Similar conclusions can be drawn from the analysis of Gen Z university students' social-networking behavior, reflecting their attitudes towards phishing awareness education [6]. Thus, the research question of this study is formulated around the role of education in improving detection capabilities against phishing.

This paper focuses on the literature review and scope definition regarding the role of education in social media phishing awareness. Research questions RQ1, RQ2, and RQ3 guide the study, while specific attention is paid to outlining the research framework for RQ1 and RQ2. Thus, RQ1 aims to identify the social-media and human factors related to phishing susceptibility, and RQ2 explores the evidence on awareness education's effectiveness. Finally, RQ3 is a follow-up question about the measures to evaluate awareness education in the context of social media use.

RQ1: Which social-media and human factors are related to phishing susceptibility?

RQ2: What is the evidence about awareness education's effectiveness?

RQ3: How can awareness education be evaluated in the context of social media use with which indicators?

3. Literature Survey

The reviewed sources were grouped into three themes instead of being presented individually. Table 1 shows an overview.

3.1 Platform context and human susceptibility

Phishing is a social engineering threat that targets users to disclose confidential information and engage with deceptive websites or malicious payloads. Nasution et al. observe that phishing attacks on Instagram are enabled by the existence of deceptive web domains and social engineering markers [2]. Of particular note, recent vignette studies demonstrate that context, including the relationship to the sender of the message on social media, is a critical predictor of susceptibility to phishing; in one such study that involved 193 young adults, perceived similarity to the message sender was responsible for the variance in susceptibility, as was trait fear of missing out [7]. Qahri-Saremi and Turel adopt a dual-process approach that conceptualizes susceptibility as the outcome of temptation and control process and identify several situational moderators of these processes in a set of five experiments, including sleep deprivation, social media ostracism, source attractiveness, and fear appeals [8]. Vardhan et al. investigate the "DM-Me" scam on Instagram and identify the variation in vulnerability to the attack, both in terms of differing scenarios and user demographics [9].

Albladi and Weir examine the influence of network engagement, motivation, experience, trust, and competence on users' ability to evaluate phishing attempts [4]; they later build upon their model to develop a means of identifying vulnerable users and targeting organizational security training [5]. In the theoretical domain, Zhuo et al. propose a comprehensive categorization of variables associated with phishing susceptibility that is organized in a conceptual model that incorporates individual, situational, and technological factors [10]. Cognitive models also contribute to an understanding of phishing susceptibility and response to security training; these include a systematic review of human and cognitive factors related to phishing detection [11], an eye-tracking study of phishing email processing [12], and the investigation of metacognitive processes in phishing detection and response [13]. Concepts from persuasion research also inform the current understanding of social media phishing, as demonstrated by Ferreira et al. in their study that combines Cialdini's principles of persuasion, Gragg's psychological triggers, and the principles of deception from Stajano et al.'s computer-supported social interactions [14]; their analysis reveals that phishing emails incorporate persuasion tactics to manipulate victims. Together, these studies demonstrate that different factors contribute to susceptibility to social media phishing that vary both at the level of individual users and their contextual circumstances; these factors are evaluated using different

methodological approaches, and, overall, few studies relate phishing susceptibility to social media messages.

3.2 Effectiveness of awareness training

The literature on the efficacy of awareness training is heterogeneous, with some studies illustrating the benefits of properly designed interventions while others reveal the limitations of traditional approaches. Sharevski and Jachim's 120-participant experimental study provides evidence of the effectiveness of interactive training delivered through voice assistants, compared to the conventional facts-and-advice approach; the results show that interactive training led to greater resistance to phishing emails, particularly in terms of the key persuasion factors [15]. Adamu et al. report on the effectiveness of an adaptive awareness training framework that utilized interactive phishing simulations in a within-subject design and found that such training increased phishing detection accuracy by $p < 0.001$ (Cohen's $d = 1.47$) and facilitated knowledge retention at 2-week follow-up [16]. Their earlier laboratory study also found embedded training, which became active when the participant engaged in deceptive behavior, to be more effective compared to standard awareness training [17].

In contrast, Ho et al. analyze an awareness campaign, which included training videos, targeted at over 19,500 employees and lasting eight months; they conclude that training and embedded deception awareness failed to provide enduring protection against phishing, as evidenced by the low detection rates at baseline and the insignificant differences across the groups at 12-month follow-up [18]. Similarly, Lain et al. examined the impact of embedded training simulations on knowledge and behavior in a study involving over 14,000 employees at 15-month follow-up and found no evidence of increased resilience to deception [19]. Training awareness was also counterproductive in this study, as it drove employees to report suspicious emails to the security team, fulfilling the attackers' objective of gathering information on network traffic. Brehmer et al. offer a similar observation on the overall limitations of awareness training, suggesting that it requires novel engagement tactics to promote behavioral change [20]. Finally, a review paper by Saini et al. examines the effectiveness of anti-phishing training, noting that the evidence is consistent in several areas while the findings on the impact of age, for example, are contradictory; the authors recommend initial organizational training combined with periodic embedded simulation targeting individual users [21]. Accordingly, the inconsistency in findings on the effectiveness of awareness training can be explained by differences in training design (interactive vs. traditional approaches), personalization (general vs. individualized feedback), sample size, study setting (lab vs. field experiment), and outcome measures (knowledge vs. behavioral outcomes).

3.3 Technical detection and emerging threats

Machine-learning detectors can classify suspicious URLs and message features and comparative work evaluates models on real phishing datasets by content and feature type [22]. Surveys of web phishing detection catalog the available techniques [23] and recent reviews examine phishing taxonomies and detection strategies, including federated learning [24]. Generative AI is now also used to produce more convincing, personalized phishing and social-engineering content [25], which undermines cues that training traditionally teaches, such as poor grammar. Technical and human defences are therefore complementary: automation reduces exposure but the user is in the decision loop.

Table 1. Summary of key studies informing the review.

Source	Context	Main contribution	Implication
Alkhalil et al. [1]	Phishing overview	Anatomy of phishing: phases, attacker types, mediums, techniques	Frame scenarios by attack phase

Nasution et al. [2]	Instagram phishing	Maps fake-site and social-engineering techniques	Use platform-specific scenarios
Jagatic et al. [3]	Social context (university students)	Lures from apparent acquaintances drew about 72% response	Include impersonated-friend lures
Albladi & Weir [4], [5]	Social networks	Links user characteristics to vulnerability	Record user factors
Zhuo et al. [10]	Human-centred phishing	Susceptibility model across person, situation, technology	Include contextual variables
Klütsch et al. [7]	Instagram messages (n = 193)	Sender relationship and fear of missing out affect susceptibility	Vary sender familiarity
Qahri-Saremi & Turel [8]	Social-media experiments	Temptation–restraint model with situational contingencies	Record situational factors
Vardhan et al. [9]	Instagram DM scams	Susceptibility varies by scam narrative and demographics	Include DM-based lures
Ferreira et al. [14]	Persuasion in phishing	Merged taxonomy of persuasion principles	Tag each scenario with its triggers
Sharevski & Jachim [15]	Training (n = 120)	Interactive training improved detection for some cues	Test interactive education
Adamu et al. [16]	Adaptive training (n = 60)	Pre/post gains and two-week retention	Add delayed retention test
Ho et al. [18]	Enterprise (n > 19,500)	Conventional training showed limited practical effect	Measure behaviour, not completion
Lain et al. [19]	Enterprise (n > 14,000; 15 months)	Embedded training did not improve resilience; reporting was practical	Record reporting as well as clicks
Schmitt & Flechais [25]	Generative AI	AI-enabled social engineering and phishing	Update scenarios regularly

4. Challenges and Research Gaps

• Limited evidence on platforms: most training studies are conducted via email [21], while only a few recent studies examine Instagram messages or scams via DM [7], [9], though these examine susceptibility rather than effect of training and few test fake support accounts, giveaways or impersonated friends within a training design

• Conflicting results: small controlled studies report large gains [15], [16] while large field trials yield little or no effect [18], [19]; differences in setting and measures prevent direct comparison

• Completion versus behaviour: many programs are evaluated in terms of attendance or quiz scores rather than unsafe-click rates in realistic scenarios

• Short follow up: retention beyond a few weeks is rarely measured, making it unclear if any gain is durable

• No common evaluation protocol: studies use different samples, scenarios and metrics, precluding a pooled estimate of effect

• Moving threat landscape: interfaces and attacker tactics (including generative AI [25]) change faster than training content

5. Proposed Methodology

To address these, we propose a mixed-method pre-test / post-test evaluation for awareness education of social-media phishing (Figure 1). Scenario-based assessment captures actual decisions, questionnaires captures knowledge and confidence, and a delayed assessment captures retention.

Figure 1. Proposed research workflow.

5.1 Design, participants and scenarios

The five-stage protocol includes (1) pre-awareness survey and recognition test; (2) controlled phishing simulation; (3) awareness education; (4) post-education simulation and survey; and (5) retention assessment after ~2 weeks. An appropriate number of participants would be 60-120 student or young-adult users divided into an intervention and possibly comparison group if ethically possible. Frequency of usage, prior security education, confidence and prior exposure were recorded as explanatory variables.

Realistic but harmless scenarios were used and classified into four archetypes (Table 3): fake platform security alert, giveaway or prize claim, impersonated friend and fake customer support. Shortened links and requests for verification codes were included as cues to these archetypes. Each was associated with the persuasion principles it exploited [14], and the observable cues to which a careful user could respond (urgency, authority, scarcity, and presence of unusual URLs, and account details). Each archetype was matched with a corresponding legitimate message so that true negatives and false positives could be scored. The response (identify, ignore, report or proceed) was recorded.

Table 3. Simulated social-media phishing scenario archetypes (fictitious accounts and domains only).

Archetype

Lure design

Psychological trigger

Observable deception cues

A1. Fake platform security alert

A "Support Team" message reports an unusual login and warns that the account will be restricted within 24 hours unless identity is confirmed by a link.

Authority; urgency; fear of loss

Handle imitates the official name with extra characters; no verified badge; shortened or look-alike URL; asks for password or one-time code; generic greeting; deadline threat.

A2. Giveaway or prize claim

A brand-style account announces winners and asks the user to "DM to claim" and confirm delivery details or pay a small fee.

Reward; scarcity; social proof from supportive comments

Recently created account with few posts; engagement inconsistent with followers; similar-looking commenter accounts; move to DM [9]; payment or personal-data request; subtly misspelled brand name.

A3. Impersonated friend

A message from a cloned or hijacked contact asks "Is this you in this video?" or asks the user to forward a code just received, with a link.

Liking and familiarity [7]; reciprocity; curiosity

Tone or timing out of character; generic wording; link opens a login look-alike page; verification-code request; username or photo differs slightly from the real contact.

A4. Fake customer support or collaboration offer

After a public complaint or brand tag, an account posing as "support" or a "partnership manager" offers help or paid collaboration and sends a form link.

Authority; helpfulness and commitment; opportunity

Unsolicited contact; handle with numerals or underscores; asks to continue on another messaging app; form on a non-official domain; asks for credentials or code screenshots; pressure to reply quickly.

5.2 Awareness education

Education combines short explanations with examples and interactive practice, emphasizing decision processes over memorization: verify the sender, inspect URLs before opening, recognize urgency and authority cues, never share credentials or one-time codes, confirm through an independent channel, use platform reporting and enable multi-factor authentication. Following the recommendation for recurring, user-specific training [21], the design allows a repeat session for participants who perform poorly.\

5.3 Evaluation measures and analysis

Let TP be phishing scenarios correctly identified, TN legitimate scenarios correctly accepted, FP legitimate scenarios wrongly flagged and FN phishing scenarios wrongly accepted:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

$$Precision = TP / (TP + FP) \quad (2)$$

$$Recall = TP / (TP + FN) \quad (3)$$

$$F1 = 2 \times Precision \times Recall / (Precision + Recall) \quad (4)$$

The unsafe-action rate is the proportion of phishing scenarios in which the participant clicks, replies, shares information or proceeds without verification. Improvement is $\Delta A = A_post - A_pre$ for the chosen metric. Pre/post scores are compared with a suitable paired test, between-group differences with an independent-samples or non-parametric test, and effect sizes are reported with p-values. Correlation or regression explores links between usage, confidence and performance, and open-ended answers are coded for why messages were trusted or rejected.

5.4 Ethical and safety controls

Simulations use fictitious domains and non-sensitive interactions; no real passwords, banking data or authentication codes are collected. Participation is voluntary, data are anonymized, and each participant receives a debriefing explaining every scenario.

6. Findings and Discussion

As a review, this paper makes no new participant-level claims. The findings below come from the reviewed literature (Table 2).

Table 2. Findings from the literature and the proposed measure.

Evidence area	Finding from reviewed literature	Measure in proposed study
Human susceptibility	User and situational characteristics influence vulnerability [4], [5], [10]	Risk-factor questionnaire + scenario performance

Social-media context	Sender relationships, fake sites and DM lures create attack opportunities [2], [3], [7], [9]	Platform-specific simulated lures (Table 3)
Persuasion cues	Phishing combines persuasion principles in specific ways [14]	Scenarios tagged by trigger
Interactive training	Improved detection of some persuasion-based phishing [15]	Pre/post scenario accuracy
Conventional and embedded training	Limited practical effect at large scale [18], [19]	Unsafe-action rate
Adaptive education	Significant improvement and two-week retention [16]	Immediate and delayed post-test

Major findings. (i) Vulnerability is distributed unevenly across users, messages, and scenarios. (ii) The effectiveness of training depends on the mode of delivery and behavioral relevance; interactive and adaptive modes provide a benefit in controlled studies, but annual passive training and embedded training have little impact in field experiments. (iii) In social media the relationship with the sender and the context of the situation are critical, implying the need for diversified scenarios [7], [8]. (iv) The 2026 adaptive-training experiment is a valuable benchmark for the future research ($p < 0.001$, $d = 1.47$, $r = 0.72$ between engagement and performance) [16]; these numbers apply to that study and should not be combined with the current study.

Trade-offs and suitability. Interactive/adaptive design allow concentrating resources on a narrow but engaged audience and using targeted persuasion elements but require high costs in construction and maintenance. Traditional methods cost-effective at a large scale, but should not be used exclusively, given the evidence. Detection is a critical first step with low overheads, but can't replace tailored training, especially in the face of AI-powered deception. A combination of detection tools and frequent, scenario-based educational interventions is optimal. Similarly, email-based training may have limited portability to social media, calling for experiments with attention to the specific contexts of a platform.

Limitations. The student recruitment base skews young and academic; the simulated attacks cannot capture the emotional charge of a real encounter; self-reported confidence scores may not reflect actual behavioral change; the heterogeneity of populations, platforms, and contexts across studies prevent aggregated analysis.

7. Conclusion

Phishing on social media is a socio-technical problem that involves both characteristics of the platform and behavioral patterns of individual users. The reviewed literature reveals that vulnerability to phishing is unevenly distributed among various factors, and the effect of different training methods differ significantly. While interactive and adaptive training methods showed some benefits in controlled studies, conventional and embedded training have little impact in field experiments. Furthermore, the reviewed literature highlights several areas for future research, including the need for platform-specific, behavior-centric, and retention-focused evaluations. As a result, future studies should adopt the framework proposed in this paper, which uses a pre/post testing design, behavioral simulations centered around four scenario prototypes, a delayed retention assessment, and the accuracy, F1-score, and unsafe-action rate as metrics. Finally, future studies should use this framework within a formal participant study and evaluate the effectiveness of repeated, adaptive, and platform-specific training in promoting sustainable behavioral change.